



Directory Services and Network Management

IE3020

3rd Year, 2nd Semester

Implementation of a Linux-based Domain Controller with Active Directory and Zabbix Monitoring

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Declaration

I certify that this report does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any university, and to the best of my knowledge and belief, it does not contain any material previously published or written by another person, except where due reference is made in text.

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1. Introduction

In modern enterprise environments, centralized user authentication, secure access control, and effective network monitoring are critical to ensuring the stability and security of the IT infrastructure. This project focuses on implementing a Linux-based solution to emulate an Active Directory (AD) environment using Samba, enabling centralized domain services without relying on proprietary Microsoft systems. Additionally, the integration of Zabbix, a powerful open-source network monitoring platform, provides real-time insights into the health and performance of network components.

The assignment involves setting up a Samba-based domain controller on an Ubuntu 20.04 server to provide Active Directory functionalities such as domain authentication, user management, and DNS services. A Fedora 29 client is configured to join this domain, demonstrating cross-platform interoperability. To complement this, Zabbix is installed and configured to monitor both the server and the client, including the setup of monitoring templates for efficient system oversight.

This project not only simulates a realistic enterprise scenario but also strengthens practical understanding of the ISO Network Management Framework by covering essential aspects such as configuration, monitoring, testing, and troubleshooting.

2. Project Objectives

The primary objective of this project is to design and implement a practical, Linux-based solution for centralized network authentication and monitoring by combining Samba Active Directory services with Zabbix monitoring tools. The specific goals in this project are,

Implement a Samba-based Active Directory Domain Controller on a Linux server, enabling centralized authentication, user account management, and domain services similar to Microsoft AD.

Configure a client machine (Fedora 29) to successfully join the Samba domain and authenticate users using Active Directory credentials.

Install and configure Zabbix Server on the domain controller for proactive monitoring of system health, resource usage, and service availability.

Deploy Zabbix Agents on both the server and client systems, enabling detailed performance data collection and alert generation.

3. System Requirements and Environment

3.1. Server Environment

Download the ISO from the official Ubuntu website and create a Virtual Machine using VMware. After following the on-screen instructions and install Ubuntu 20.04 LTS.

- CPU: Minimum 2 vCPUs
- Memory: Minimum 4GB RAM (8 GB recommended for Zabbix + Samba)
- Storage: Minimum 50 GB available disk space
- Network: Static IP configuration with internet access
- Choose Bridged Network or NAT with port forwarding

1. Set hostname

A static IP address was configured, and the hostname was set to ensure consistent network identification, forming the foundation for domain controller setup and reliable network communication.

Sudo hostnamectl set-hostname dc.hewapathirana.sa

```
sawera@ubuntu:~$ hostname -f  
dc.hewapathirana.sa  
sawera@ubuntu:~$ sudo apt update && sudo apt upgrade -y
```

2. Update System Packages

`sudo apt update && sudo apt upgrade -y`

This is used to keep the system up to date with the latest security patches, bug fixes, and features by updating and upgrading all software packages.

```
root@dc:~# sudo apt update && sudo apt upgrade -y
Hit:1 http://1k.archive.ubuntu.com/ubuntu focal InRelease
Get:2 http://1k.archive.ubuntu.com/ubuntu focal-updates InRelease [128 kB]
Hit:3 http://1k.archive.ubuntu.com/ubuntu focal-backports InRelease
Get:4 http://1k.archive.ubuntu.com/ubuntu focal-security InRelease [128 kB]
Get:5 http://1k.archive.ubuntu.com/ubuntu focal-updates/main amd64 Packages [3,863 kB]
Get:6 http://1k.archive.ubuntu.com/ubuntu focal-updates/universe amd64 Packages [1,265 kB]
Get:7 http://1k.archive.ubuntu.com/ubuntu focal-security/main amd64 Packages [3,474 kB]
76% [6 Packages store 0 B] [7 Packages 1,616 kB/3,474 kB 47%]
```

3. Set time zone

Time zone is very important, especially when dealing with Kerberos authentication, which is time-sensitive

```
sawera@ubuntu:~$ timedatectl
          Local time: Tue 2025-04-08 13:49:31 PKT
          Universal time: Tue 2025-04-08 08:49:31 UTC
            RTC time: Tue 2025-04-08 08:49:31
            Time zone: Asia/Karachi (PKT, +0500)
System clock synchronized: yes
          NTP service: active
          RTC in local TZ: no
sawera@ubuntu:~$ sudo timedatectl set-timezone Asia/Colombo
sawera@ubuntu:~$ timedatectl
          Local time: Tue 2025-04-08 14:23:18 +0530
          Universal time: Tue 2025-04-08 08:53:18 UTC
            RTC time: Tue 2025-04-08 08:53:19
            Time zone: Asia/Colombo (+0530, +0530)
System clock synchronized: yes
          NTP service: active
          RTC in local TZ: no
sawera@ubuntu:~$ _
```

4. Edit /etc/hosts File

The file is used to map IP addresses to hostnames locally, without relying on DNS.

```
Vi /etc/hosts
```

```
127.0.0.1 localhost
127.0.1.1 dc.hewapathirana.sa dc
192.168.101.162 dc.hewapathirana.sa dc

# The following lines are desirable for IPv6 capable hosts
::1          ip6-localhost ip6-loopback
fe00::0      ip6-localnet
ff00::0      ip6-mcastprefix
ff02::1      ip6-allnodes
ff02::2      ip6-allrouters
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
-- INSERT --
```

2,33

All

3.2. Client Environment

Install Fedora on VMware and set up.

- Operating System: Fedora Linux 29 (Workstation Edition)
- CPU: Minimum 2 vCPUs
- Memory: Minimum 2 GB RAM
- Storage: Minimum 20 GB available disk space
- Network: DHCP or Static IP (must communicate with the domain controller and Zabbix server)

Prepare Fedora 29 to:

- Join the Samba AD domain
- Authenticate using AD credentials
- Be monitored by Zabbix Server

1. Set hostname

`Sudo hostnamectl set-hostname client1.hewapathirana.sa`

```
[root@csa-client ~]# hostnamectl set-hostname client1.hewapathirana.sa
[root@csa-client ~]# hostname
client1.hewapathirana.sa
[root@csa-client ~]# vi /etc/resolv.conf
[root@csa-client ~]#
```

2. Edit /etc/resolv.conf file

The /etc/resolv.conf file on the Fedora client was configured to use the domain controller's IP address as the primary DNS server, ensuring proper name resolution and successful domain discovery during the domain join process.

Vi /etc/resolv.conf

```
root@csa-client:~  
File Edit View Search Terminal Help  
# Generated by NetworkManager  
search localdomain hewapathirana.sa  
nameserver 192.168.101.162  
~  
~  
~  
~  
~  
~  
~
```

3. Find the connection name and set the static IP

```
root@csa-client:~  
File Edit View Search Terminal Help  
[root@csa-client ~]# nmcli con show  
NAME      UUID                                  TYPE      DEVICE  
ens160    40275112-e927-329d-aec1-38d886ef26c1 ethernet  ens160  
virbr0    874bb644-3c2e-416f-b446-437c100b0737 bridge    virbr0  
ens224    cd8f6cfb-c4b5-3a5c-b133-1ed1c2d66bed ethernet  --  
[root@csa-client ~]# nmcli con mod "ens160" ipv4.dns "192.168.101.162"  
[root@csa-client ~]# nmcli con mod 'ens160' ipv4.ig yes  
Error: invalid property 'ig': 'ig' is ambiguous: ignore-auto-routes, ignore-auto-dns.  
[root@csa-client ~]# nmcli con mod "ens160" ipv4.ignore-auto-dns yes  
Error: value for 'ipv4.ignore-auto-dns yes' is missing.  
[root@csa-client ~]# nmcli con mod "ens160" ipv4.ignore-auto-dns  
Error: value for 'ipv4.ignore-auto-dns' is missing.  
[root@csa-client ~]# nmcli con mod "ens160" ipv4.ignore-auto-dns YES  
[root@csa-client ~]#
```

network and DNS configuration steps on a Linux server. confirm that the network interface is up and using the correct DNS settings, which is important for services like Active Directory authentication.

```
root@csa-client:~  
File Edit View Search Terminal Help  
[root@csa-client ~]# nmcli con down "ens160"  
Connection 'ens160' successfully deactivated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/1)  
[root@csa-client ~]# nmcli con up "ens160"  
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/8)  
[root@csa-client ~]# nmcli dev show | grep DNS  
IP4.DNS[1]: 192.168.101.162  
[root@csa-client ~]# cat /etc/resolv.conf  
# Generated by NetworkManager  
search hewapathirana.sa  
nameserver 192.168.101.162  
[root@csa-client ~]#
```

4. Implementation Details

4.1. Setting up Samba Active Directory Domain Controller

1. Install Required Packages

`sudo apt install -y samba krb5-config krb5-user winbind smbclient dnsutils`

```
root@dc:~# sudo apt install -y samba krb5-config krb5-user winbind smbclient dnsutils
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  attr ibverbs-providers libavahi-client3 libavahi-common-data libavahi-common3
  libboost-iostreams1.71.0 libboost-thread1.71.0 libcephfs2 libcups2 libgssrpc4 libibverbs1
  libkadm5clnt-mit11 libkadm5srv-mit11 libkdb5-9 libldb2 librados2 librdmacm1 libsmbclient
  libtalloc2 libtevent0 libwbclient0 python3-crypto python3-dnspython python3-gpg python3-ldb
  python3-markdown python3-pygments python3-samba python3-talloc python3-tdb samba-common
  samba-common-bin samba-dsdb-modules samba-libs samba-vfs-modules tdb-tools
Suggested packages:
  krb5-k5tls cups-common krb5-doc python-markdown-doc python-pygments-doc ttf-bitstream-vera bind9
  bind9utils ctdb ldb-tools ntp | chrony smbldap-tools heimdal-clients cifs-utils libnss-winbind
  libpam-winbind
The following NEW packages will be installed:
  attr dnsutils ibverbs-providers krb5-config krb5-user libavahi-client3 libavahi-common-data
  libavahi-common3 libboost-iostreams1.71.0 libboost-thread1.71.0 libcephfs2 libcups2 libgssrpc4
  libibverbs1 libkadm5clnt-mit11 libkadm5srv-mit11 libkdb5-9 libldb2 librados2 librdmacm1
  libsmbclient libtalloc2 libtevent0 libwbclient0 python3-crypto python3-dnspython python3-gpg
  python3-ldb python3-markdown python3-pygments python3-samba python3-talloc python3-tdb samba
  samba-common samba-common-bin samba-dsdb-modules samba-libs samba-vfs-modules smbclient
  tdb-tools winbind
0 upgraded, 42 newly installed, 0 to remove and 0 not upgraded.
Need to get 18.6 MB of archives.
After this operation, 108 MB of additional disk space will be used.
0% [Waiting for headers]
```

Kerberos Realm

Configuring Kerberos Authentication

When users attempt to use Kerberos and specify a principal or user name without specifying what administrative Kerberos realm that principal belongs to, the system appends the default realm. The default realm may also be used as the realm of a Kerberos service running on the local machine. Often, the default realm is the uppercase version of the local DNS domain.

Default Kerberos version 5 realm:

HEWAPATHIRANA.SA

<Ok>

Kerberos Server

Package configuration

Configuring Kerberos Authentication

Enter the hostnames of Kerberos servers in the HEWAPATHIRANA.SA Kerberos realm separated by spaces.
Kerberos servers for your realm:
dc.hewapathirana.sa
<Ok>

Admin Sever

Package configuration

Configuring Kerberos Authentication

Enter the hostname of the administrative (password changing) server for the HEWAPATHIRANA.SA Kerberos realm.
Administrative server for your Kerberos realm:
dc.hewapathirana.sa
<Ok>

2. Provision the Domain Controller

Backup Existing Configurations

```
root@dc:~# sudo mv /etc/samba/smb.conf /etc/samba/smb.conf.bak
root@dc:~# _
```

Run the Samba provisioning tool interactively

```
root@dc:~# sudo samba-tool domain provision --use-rfc2307 --interactive
Realm [HEWAPATHIRANA.SA]: HEWAPATHIRANA.SA
Domain [HEWAPATHIRANA]: HEWAPATHIRANA
Server Role (dc, member, standalone) [dc]: dc
DNS backend (SAMBA_INTERNAL, BIND9_FLATFILE, BIND9_DLZ, NONE) [SAMBA_INTERNAL]: SAMBA_INTERNAL
DNS forwarder IP address (write 'none' to disable forwarding) [127.0.0.53]: 8.8.8.8
Administrator password:
```

3. Configure DNS and Kerberos

Vi /etc/krb5.conf

Updated /etc/krb5.conf to reflect the correct realm and domain settings.

```
[libdefaults]
    default_realm = HEWAPATHIRANA.SA
    dns_lookup_realm = false
    dns_lookup_kdc = true

# The following krb5.conf variables are only for MIT Kerberos.
    kdc_timesync = 1
    ccache_type = 4
    forwardable = true
    proxiable = true

# The following encryption type specification will be used by MIT Kerberos
# if uncommented. In general, the defaults in the MIT Kerberos code are
# correct and overriding these specifications only serves to disable new
# encryption types as they are added, creating interoperability problems.
#
# The only time when you might need to uncomment these lines and change
# the enctype is if you have local software that will break on ticket
# caches containing ticket encryption types it doesn't know about (such as
# old versions of Sun Java).
#
    default_tgs_enctypes = des3-hmac-sha1
    default_tkt_enctypes = des3-hmac-sha1
    permitted_enctypes = des3-hmac-sha1

# The following libdefaults parameters are only for Heimdal Kerberos.
    fcc-mit-ticketflags = true

[realms]
    HEWAPATHIRANA.SA = {
        kdc = dc.hewapathirana.sa
        admin_server = dc.hewapathirana.sa
    }
    ATHENA.MIT.EDU = {
        kdc = kerberos.mit.edu
        kdc = kerberos-1.mit.edu
    }

:wq!
```

4. Start Samba service

Samba has three main services:

- `smbd`: Handles file sharing and authentication.
- `nmbd`: Handles NetBIOS name resolution (optional with AD).
- `Winbind`: used for Windows authentication.

To run Samba in Active Directory Domain Controller (AD DC) mode, first stop the traditional Samba services (`smbd`, `nmbd`, and `winbind`) as they are replaced by the `samba-ad-dc` service. Then, unmask the `samba-ad-dc` service if it's disabled by default, enable it to start on boot, and start it immediately to activate AD DC functionality.

```
root@dc:~# sudo systemctl stop smbd nmbd winbind
root@dc:~# sudo systemctl unmask samba-ad-dc
Removed /etc/systemd/system/samba-ad-dc.service.
root@dc:~# sudo systemctl enable samba-ad-dc
Synchronizing state of samba-ad-dc.service with SysV service script with /lib/systemd/systemd-sysv-in
stall.
Executing: /lib/systemd/systemd-sysv-install enable samba-ad-dc
Created symlink /etc/systemd/system/multi-user.target.wants/samba-ad-dc.service → /lib/systemd/syste
m/samba-ad-dc.service.
root@dc:~# sudo systemctl start samba-ad-dc
root@dc:~#
```

Sudo systemctl status samba-ad-dc

Verify that the service is running correctly. look for active (running).

```
• samba-ad-dc.service - Samba AD Daemon
   Loaded: loaded (/lib/systemd/system/samba-ad-dc.service; enabled; vendor preset: enabled)
   Active: active (running) since Sat 2025-04-12 11:52:23 +0530; 44s ago
     Docs: man:samba(8)
           man:samba(7)
           man:smb.conf(5)
  Main PID: 7194 (samba)
    Status: "samba: ready to serve connections..."
     Tasks: 59 (limit: 4554)
    Memory: 207.2M
   CGroup: /system.slice/samba-ad-dc.service
           └─7194 samba: root process
             └─7207 samba: tfork waiter process(7208)
               └─7208 samba: task[s3fs] pre-fork master
                 └─7209 samba: tfork waiter process(7211)
                   └─7210 samba: tfork waiter process(7212)
                     └─7211 samba: task[rpc] pre-fork master
                       └─7212 /usr/sbin/smbd -D --option=server role check:inhibit=yes --foreground
                         └─7213 samba: tfork waiter process(7214)
                           └─7214 samba: task[nbt] pre-fork master
                             └─7215 samba: tfork waiter process(7216)
                               └─7216 samba: task[wrepl] pre-fork master
                                 └─7217 samba: tfork waiter process(7219)
                                   └─7218 samba: tfork waiter process(7221)
                                     └─7219 samba: task[ldap] pre-fork master
                                       └─7220 samba: tfork waiter process(7223)
                                         └─7221 samba: task[rpc] pre-forked worker(0)
                                           └─7222 samba: tfork waiter process(7224)
                                             └─7223 samba: task[cldap] pre-fork master
                                               └─7224 samba: task[rpc] pre-forked worker(1)
                                                 └─7225 samba: tfork waiter process(7227)
                                                   └─7226 samba: tfork waiter process(7229)
                                                     └─7227 samba: task[rpc] pre-forked worker(2)
                                                       └─7228 samba: tfork waiter process(7230)
                                                         └─7229 samba: task[kdc] pre-fork master
                                                           └─7230 samba: task[rpc] pre-forked worker(3)
```

lines 1-36

5. Verifying Active Directory DNS and Authentication

Check that your Ubuntu server can resolve the AD domain and domain controllers via DNS

```
root@dc:~# host -t SRV _ldap._tcp.hewapathirana.sa
Host _ldap._tcp.hewapathirana.sa not found: 3(NXDOMAIN)
root@dc:~# _
```

Restarts the Samba Active Directory Domain Controller (AD DC) service. Restarting ensures that any configuration changes to Samba or the system are applied, and the service is running with the latest settings.

```
root@dc:~# systemctl restart samba-ad-dc.service
root@dc:~# host -t SRV _ldap._tcp.hewapathirana.sa
_ldap._tcp.hewapathirana.sa has SRV record 0 100 389 dc.hewapathirana.sa.
root@dc:~#
```

Test with kinit

Kerberos is a key authentication protocol used by Active Directory for securely validating user identities

If authentication is successful, the command will return silently.

```
root@dc:~# host -t SRV _kerberos._udp.hewapathirana.sa
_kerberos._udp.hewapathirana.sa has SRV record 0 100 88 dc.hewapathirana.sa.
root@dc:~# kinit administrator
Password for administrator@HEWAPATHIRANA.SA:
```

Test Samba authentication by creating a test user

To test Samba AD authentication, you can create a test user in your domain using the samba-tool utility, then attempt to authenticate that user.

```
root@dc:~# sudo samba-tool user add testuser
New Password:
Retype Password:
User 'testuser' added successfully
root@dc:~# _
```

```
root@dc:~# sudo samba-tool user list
krbtgt
Guest
Administrator
testuser
root@dc:~# _
```

```
testuser
root@dc:~# kinit testuser@HEWAPATHIRANA.SA
Password for testuser@HEWAPATHIRANA.SA:
Warning: Your password will expire in 41 days on Sat 24 May 2025 12:39:25 PM +0530
root@dc:~#
```

Verify the Ticket with klist

This confirms that the Kerberos ticket was issued successfully and the system can authenticate users via Kerberos.

```
root@dc:~# klist
Ticket cache: FILE:/tmp/krb5cc_0
Default principal: testuser@HEWAPATHIRANA.SA

Valid starting    Expires          Service principal
04/12/2025 12:42:49 04/12/2025 22:42:49 krbtgt/HEWAPATHIRANA.SA@HEWAPATHIRANA.SA
        renew until 04/13/2025 12:42:40
root@dc:~#
```

4.2. Installing and Configuring Zabbix Server

1. Install Zabbix Repository and Server Packages

Zabbix provides its own package repository. Add the Zabbix repository to get the latest stable Zabbix packages for Ubuntu.

```
root@dc:~# wget https://repo.zabbix.com/zabbix/6.0/ubuntu/pool/main/z/zabbix-release/zabbix-release_
latest_6.0+ubuntu20.04_all.deb
--2025-05-04 11:33:01-- https://repo.zabbix.com/zabbix/6.0/ubuntu/pool/main/z/zabbix-release/zabbix
-release_latest_6.0+ubuntu20.04_all.deb
Resolving repo.zabbix.com (repo.zabbix.com)... 178.128.6.101, 2604:a880:2:d0::2062:d001
Connecting to repo.zabbix.com (repo.zabbix.com)|178.128.6.101|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 3672 (3.6K) [application/octet-stream]
Saving to: 'zabbix-release_latest_6.0+ubuntu20.04_all.deb'

zabbix-release_latest_6. 100%[=====>] 3.59K --.-KB/s in 0s

2025-05-04 11:33:02 (226 MB/s) - 'zabbix-release_latest_6.0+ubuntu20.04_all.deb' saved [3672/3672]

root@dc:~#
```

The Debian package manager is used to install .deb packages manually. **zabbix-release_6.0+ubuntu20.04_all.deb** is the Zabbix repository package for version 6.0, specifically built for Ubuntu 20.04.

```
root@dc:~# dpkg -i zabbix-release_latest_6.0+ubuntu20.04_all.deb
Selecting previously unselected package zabbix-release.
(Reading database ... 74825 files and directories currently installed.)
Preparing to unpack zabbix-release_latest_6.0+ubuntu20.04_all.deb ...
Unpacking zabbix-release (1:6.0-4+ubuntu20.04) ...
Setting up zabbix-release (1:6.0-4+ubuntu20.04) ...
root@dc:~#
```

Apt update.

This tells apt to fetch the latest list of available packages from all configured repositories. It does not upgrade packages; it simply refreshes the package database, so the system knows which versions of software are currently available. Always run `sudo apt update` before installing new packages to ensure to get the latest versions

```
root@dc:~# apt update
Hit:1 http://lk.archive.ubuntu.com/ubuntu focal InRelease
Get:2 http://lk.archive.ubuntu.com/ubuntu focal-updates InRelease [128 kB]
Get:3 https://repo.zabbix.com/zabbix-agent2-plugins/1/ubuntu focal InRelease [4,952 B]
Get:4 http://lk.archive.ubuntu.com/ubuntu focal-backports InRelease [128 kB]
Get:5 https://repo.zabbix.com/zabbix/6.0/ubuntu focal InRelease [2,886 B]
Get:6 http://lk.archive.ubuntu.com/ubuntu focal-security InRelease [128 kB]
Get:7 http://lk.archive.ubuntu.com/ubuntu focal-updates/main amd64 Packages [3,921 kB]
Get:8 https://repo.zabbix.com/zabbix-agent2-plugins/1/ubuntu focal/main Sources [1,004 B]
Get:9 https://repo.zabbix.com/zabbix-agent2-plugins/1/ubuntu focal/main amd64 Packages [625 B]
Get:10 https://repo.zabbix.com/zabbix/6.0/ubuntu focal/main Sources [37.9 kB]
Get:11 http://lk.archive.ubuntu.com/ubuntu focal-updates/main Translation-en [595 kB]
Get:12 http://lk.archive.ubuntu.com/ubuntu focal-updates/restricted amd64 Packages [3,854 kB]
36% [7 Packages store 0 B] [12 Packages 0 B/3,854 kB 0%] [10 Sources 16.1 kB/37.9 kB 43%]
```

2. Install Zabbix server, frontend, agent

```
root@dc:~# apt install zabbix-server-pgsql zabbix-frontend-php php7.4-pgsql zabbix-apache-conf zabbix-sql-scripts zabbix-agent
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils fontconfig-config fonts-dejavu fonts-dejavu-core fonts-dejavu-extra fping libapache2-mod-php libapache2-mod-php7.4 libapr1 libaprutil1
  libaprutil1-dbd-sqlite3 libaprutil1-ldap libfontconfig1 libgd3 libjbig0 libjpeg-turbo8 libjpeg8 liblua5.2-0 libmodbus5 libmysqlclient21 libodbc1 libonig5 libopenipmi0 libpq5 libsensors-config
  libsensors5 libsnmp-base libsnmp35 libtiff5 libwebp6 libxpm4 mysql-common php-bcmath php-common php-gd php-ldap php-mbstring php-xml php7.4-bcmath php7.4-cli php7.4-common php7.4-gd
  php7.4-json php7.4-ldap php7.4-mbstring php7.4-opcache php7.4-readline php7.4-xml snmpd ssl-cert
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-browser php-pear libgd-tools libmyodbc odbc-postgresql tdsodbc unixodbc-bin lm-sensors snmp-mibs-downloader snmptrapd
  openssl-blacklist zabbix-nginx-conf postgresql-client postgresql
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils fontconfig-config fonts-dejavu fonts-dejavu-core fonts-dejavu-extra fping libapache2-mod-php libapache2-mod-php7.4 libapr1 libaprutil1
  libaprutil1-dbd-sqlite3 libaprutil1-ldap libfontconfig1 libgd3 libjbig0 libjpeg-turbo8 libjpeg8 liblua5.2-0 libmodbus5 libmysqlclient21 libodbc1 libonig5 libopenipmi0 libpq5 libsensors-config
  libsensors5 libsnmp-base libsnmp35 libtiff5 libwebp6 libxpm4 mysql-common php-bcmath php-common php-gd php-ldap php-mbstring php-xml php7.4-bcmath php7.4-cli php7.4-common php7.4-gd
  php7.4-json php7.4-ldap php7.4-mbstring php7.4-opcache php7.4-pgsql php7.4-readline php7.4-xml snmpd ssl-cert zabbix-agent zabbix-apache-conf zabbix-frontend-php zabbix-server-pgsql
  zabbix-sql-scripts
0 upgraded, 59 newly installed, 0 to remove and 17 not upgraded.
Need to get 29.9 MB of archives.
After this operation, 109 MB of additional disk space will be used.
Do you want to continue? [Y/n] _
```

4.3. PostgreSQL Database Configuration

Install PostgreSQL using this command: `sudo apt install -y postgresql-14`

```
root@dc:~# sudo apt install potstgresql-14 postgresql-client-14 -y
Reading package lists... 69%
```

Testing if PostgreSQL is active.

```
root@dc:~# systemctl enable postgresql
Synchronizing state of postgresql.service with SysV service script with /lib/systemd/systemd-sysv-in
stall.
Executing: /lib/systemd/systemd-sysv-install enable postgresql
root@dc:~# systemctl start postgresql
root@dc:~# systemctl status postgresql
• postgresql.service - PostgreSQL RDBMS
   Loaded: loaded (/lib/systemd/system/postgresql.service; enabled; vendor preset: enabled)
   Active: active (exited) since Sun 2025-05-04 11:53:55 +0530; 53s ago
   Main PID: 19695 (code=exited, status=0/SUCCESS)
   Tasks: 0 (limit: 4554)
   Memory: 0B
   CGroup: /system.slice/postgresql.service

May 04 11:53:55 dc.hewapathirana.sa systemd[1]: Starting PostgreSQL RDBMS...
May 04 11:53:55 dc.hewapathirana.sa systemd[1]: Finished PostgreSQL RDBMS.
root@dc:~# sudo -u postgres createuser --pwprompt zabbix
could not change directory to "/root": Permission denied
Enter password for new role:
Enter it again:
root@dc:~# sudo -u postgres createdb -O zabbix zabbix
could not change directory to "/root": Permission denied
root@dc:~# sudo -u postgres createdb -O zabbix zabbix
could not change directory to "/root": Permission denied
createdb: error: database creation failed: ERROR: database "zabbix" already exists
root@dc:~# _
```

1. Import Initial Schema

Loads the necessary tables, functions, and default data into the Zabbix database.

```
root@dc:~# zcat /usr/share/zabbix-sql-scripts/postgresql/server.sql.gz | sudo -u zabbix psql zabbix
```

2. Edit Zabbix Server Config

Vi /etc/Zabbix/Zabbix_server.conf

Update the following lines:

- DBName=Zabbix
- DBUser=Zabbix
- DBPassword=Zabbix

Ensures the Zabbix server knows how to connect to the PostgreSQL database.

```
# Database user.
#
# Mandatory: no
# Default:
# DBUser=
DBUser=zabbix

### Option: DBPassword
# Database password.
# Comment this line if no password is used.
#
# Mandatory: no
# Default:
DBPassword=zabbix

### Option: DBSocket
# Path to MySQL socket.
#
# Mandatory: no
# Default:
# DBSocket=

### Option: DBPort
# Database port when not using local socket.
# If the Net Service Name connection method is used to connect to Oracle database, the port number from the
# tnsnames.ora file will be used. The port number set here will be ignored.
#
# Mandatory: no
# Range: 1024-65535
# Default:
# DBPort=

### Option: AllowUnsupportedDBVersions
# Allow server to work with unsupported database versions.
-- INSERT --
```

120,1 11%

3. Restart and Enable Services

```
root@dc:~# systemctl restart zabbix-server zabbix-agent apache2
root@dc:~# systemctl enable zabbix-server zabbix-agent apache2
Synchronizing state of zabbix-server.service with SysV service script with /lib/systemd/systemd-sysv
-install.
Executing: /lib/systemd/systemd-sysv-install enable zabbix-server
Synchronizing state of zabbix-agent.service with SysV service script with /lib/systemd/systemd-sysv-
install.
Executing: /lib/systemd/systemd-sysv-install enable zabbix-agent
Synchronizing state of apache2.service with SysV service script with /lib/systemd/systemd-sysv-insta
ll.
Executing: /lib/systemd/systemd-sysv-install enable apache2
Created symlink /etc/systemd/system/multi-user.target.wants/zabbix-server.service → /lib/systemd/sys
tem/zabbix-server.service.
```

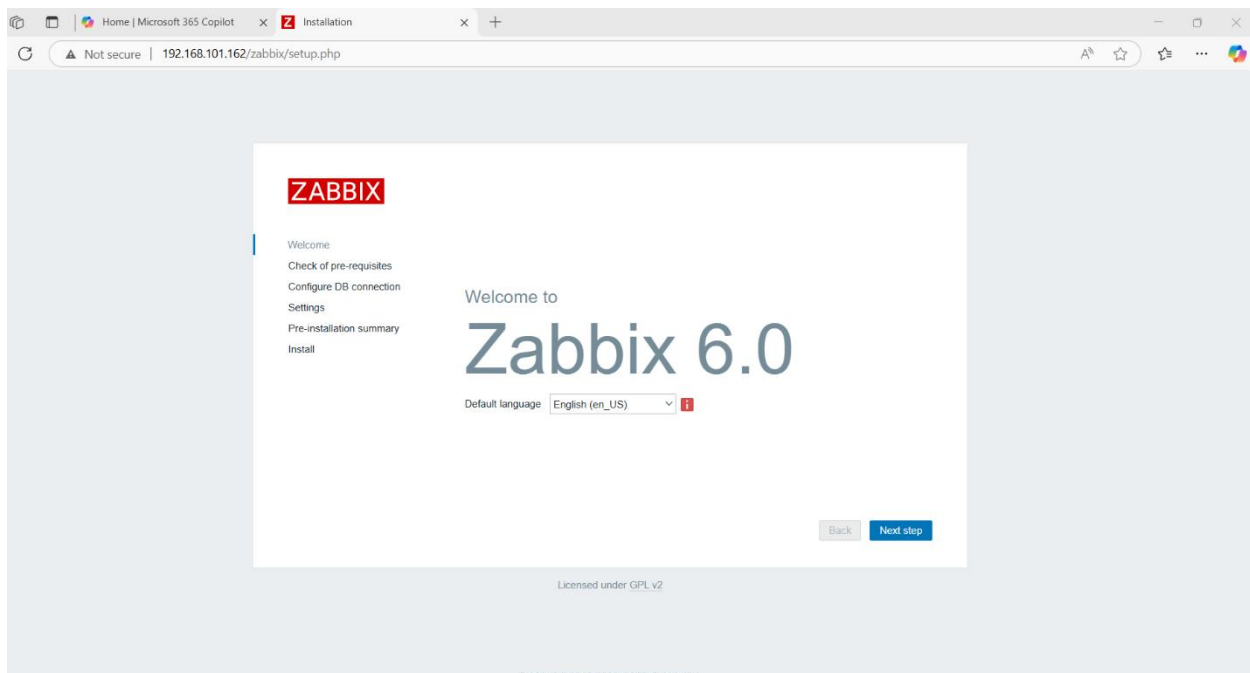
4.4. Zabbix Frontend Configuration

Open your browser and type the default URL for Zabbix UI when using the Apache web server is <http://192.168.101.162/zabbix>

This is the Zabbix web installer. If you complete the final steps via a GUI, enter DB details, time zone, and admin password.

Default login:

- **Username:** Admin
- **Password:** zabbix





Configure DB connection

Please create database manually, and set the configuration parameters for connection to this database. Press "Next step" button when done.

Welcome

Check of pre-requisites

Configure DB connection

Settings

Pre-installation summary

Install

Database type

Database host

Database port 0 - use default port

Database name

Database schema

Store credentials in ☐ Plain text ☐ HashiCorp Vault

User

Password

Database TLS encryption ☒

Back

Next step

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Settings

Zabbix server name

Default time zone

Default theme

Welcome

Check of pre-requisites

Configure DB connection

Settings

Pre-installation summary

Install

Back

Next step

4.5. Installing and Configuring Zabbix Agent on Domain Controller

The domain controller is running on **Ubuntu** and has already joined the Samba AD DC.

Verify Agent Status

`sudo systemctl status zabbix-agent`

```
• zabbix-server.service - Zabbix Server
   Loaded: loaded (/lib/systemd/system/zabbix-server.service; enabled; vendor preset: enabled)
   Active: active (running) since Sun 2025-05-04 12:06:31 +0530; 1min 5s ago
   Main PID: 20826 (zabbix_server)
     Tasks: 48 (limit: 4554)
    Memory: 50.5M
    CGroup: /system.slice/zabbix-server.service
            └─20826 /usr/sbin/zabbix_server -c /etc/zabbix/zabbix_server.conf
               └─20842 /usr/sbin/zabbix_server: ha manager
                  └─20844 /usr/sbin/zabbix_server: service manager #1 [processed 0 events, updated 0 eve
                     └─20845 /usr/sbin/zabbix_server: configuration syncer [synced configuration in 0.27569s
                        └─20866 /usr/sbin/zabbix_server: alert manager #1 [sent 0, failed 0 alerts, idle 5.020s
                           └─20867 /usr/sbin/zabbix_server: alerter #1 started
                              └─20868 /usr/sbin/zabbix_server: alerter #2 started
                                 └─20869 /usr/sbin/zabbix_server: alerter #3 started
                                    └─20870 /usr/sbin/zabbix_server: preprocessing manager #1 [queued 0, processed 3 value
                                       └─20871 /usr/sbin/zabbix_server: preprocessing worker #1 started
                                          └─20872 /usr/sbin/zabbix_server: preprocessing worker #2 started
                                             └─20873 /usr/sbin/zabbix_server: preprocessing worker #3 started
                                                └─20875 /usr/sbin/zabbix_server: lld manager #1 [processed 0 LLD rules, idle 5.019640s
                                                   └─20876 /usr/sbin/zabbix_server: lld worker #1 [processed 1 LLD rules, idle 53.379674 s
                                                      └─20878 /usr/sbin/zabbix_server: lld worker #2 started
                                                         └─20879 /usr/sbin/zabbix_server: housekeeper [startup idle for 30 minutes]
                                                            └─20880 /usr/sbin/zabbix_server: timer #1 [updated 0 hosts, suppressed 0 events in 0.0
                                                               └─20881 /usr/sbin/zabbix_server: http poller #1 [got 0 values in 0.001438 sec, idle 5 >
                                                                  └─20882 /usr/sbin/zabbix_server: discoverer #1 [processed 0 rules in 0.001208 sec, idl
                                                                     └─20883 /usr/sbin/zabbix_server: history syncer #1 [processed 0 values, 0 triggers in >
                                                                        └─20884 /usr/sbin/zabbix_server: history syncer #2 [processed 0 values, 0 triggers in >
                                                                           └─20891 /usr/sbin/zabbix_server: history syncer #3 [processed 0 values, 0 triggers in >
                                                                              └─20892 /usr/sbin/zabbix_server: history syncer #4 [processed 1 values, 1 triggers in >
                                                                                 └─20893 /usr/sbin/zabbix_server: escalator #1 [processed 0 escalations in 0.002447 sec
                                                                                     └─20894 /usr/sbin/zabbix_server: proxy poller #1 [exchanged data with 0 proxies in 0.0
                                                                                         └─20899 /usr/sbin/zabbix_server: self-monitoring [processed data in 0.000044 sec, idle
                                                                                             └─20900 /usr/sbin/zabbix_server: task manager [processed 0 task(s) in 0.000689 sec, id
                                                                                                 └─20901 /usr/sbin/zabbix_server: poller #1 [got 0 values in 0.000056 sec, idle 5 sec]
                                                                                                     └─20902 /usr/sbin/zabbix_server: poller #2 [got 0 values in 0.000015 sec, idle 5 sec]
```

lines 1-36

Apt update

The apt update command refreshes your system's local package index by downloading the latest lists of available software and security updates from the repositories you've configured. This ensures your package manager knows about any newly added, removed, or upgraded packages before you install or upgrade software. Without running apt update first, your system may not recognize the newest versions of applications even if they're already available online. By synchronizing your local cache with the remote repositories, apt update lays the groundwork for reliable, up-to-date installations and upgrades

```
root@dc:~# apt update

Hit:1 http://lk.archive.ubuntu.com/ubuntu focal InRelease
Hit:2 http://lk.archive.ubuntu.com/ubuntu focal-updates InRelease
Hit:3 http://lk.archive.ubuntu.com/ubuntu focal-backports InRelease
Hit:4 http://apt.postgresql.org/pub/repos/apt focal-pgdg InRelease
Hit:5 http://lk.archive.ubuntu.com/ubuntu focal-security InRelease
Hit:6 https://repo.zabbix.com/zabbix-agent2-plugins/1/ubuntu focal InRelease
Hit:7 https://repo.zabbix.com/zabbix/6.0/ubuntu focal InRelease
Reading package lists... Done
Building dependency tree
Reading state information... Done
17 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@dc:~#
root@dc:~# apt install zabbix-agent
Reading package lists... Done
Building dependency tree
Reading state information... Done
zabbix-agent is already the newest version (1:6.0.40-1+ubuntu20.04).
0 upgraded, 0 newly installed, 0 to remove and 17 not upgraded.
root@dc:~#
```

1. Configure the Zabbix Agent

Vi /etc/zabbix/zabbix_agentd.conf

The following lines are correctly set:

- Server: IP of the Zabbix server (192.168.101.162).
- ServerActive: IP of the Zabbix server (192.1068.101.162).
- Hostname: Must match what is entered in the Zabbix frontend (hewapathirana.sa)

```
# ServerActive=127.0.0.1:20051,zabbix.domain,[:,1]:30051,::1,[12fc::1]
# Example for high availability:
# ServerActive=zabbix.cluster.node1;zabbix.cluster.node2:20051;zabbix.cluster.node3
# Example for high availability with two clusters and one server:
# ServerActive=zabbix.cluster.node1;zabbix.cluster.node2:20051,zabbix.cluster2.node1;zabbix.cluster2.node2,zabbix.domain
#
# Mandatory: no
# Default:
# ServerActive=

ServerActive=127.0.0.1,192.168.101.162

### Option: Hostname
# List of comma delimited unique, case sensitive hostnames.
# Required for active checks and must match hostnames as configured on the server.
# Value is acquired from HostnameItem if undefined.
#
# Mandatory: no
# Default:
# Hostname=

Hostname=dc.hewapathirana.sa

### Option: HostnameItem
# Item used for generating Hostname if it is undefined. Ignored if Hostname is defined.
# Does not support UserParameters or aliases.
#
# Mandatory: no
# Default:
# HostnameItem=system.hostname

### Option: HostMetadata
# Optional parameter that defines host metadata.
# Host metadata is used at host auto-registration process.
# An agent will issue an error and not start if the value is over limit of 255 characters.
-- INSERT --
```

183,29 31%

2. Start and Enable the Agent and Verify Agent Status

```
sudo systemctl restart zabbix-agent
```

```
sudo systemctl enable zabbix-agent
```

Ensure the agent starts now and on every boot.

```
sudo systemctl status zabbix-agent
```

```
# Default:
# HostnameItem=system.hostname

### Option: HostMetadata
#   Optional parameter that defines host metadata.
#   Host metadata is used at host auto-registration process.
#   An agent will issue an error and not start if the value is over limit of 255 characters.
"/etc/zabbix/zabbix_agentd.conf" 545L, 17015C written
root@dc:~# systemctl restart zabbix-
zabbix-agent.service    zabbix-server.service
root@dc:~# systemctl restart zabbix-agent.service
root@dc:~# systemctl enable zabbix-agent.service
Synchronizing state of zabbix-agent.service with SysV service script with /lib/systemd/systemd-sysv-
install.
Executing: /lib/systemd/systemd-sysv-install enable zabbix-agent
root@dc:~# systemctl status zabbix-agent.service
• zabbix-agent.service - Zabbix Agent
   Loaded: loaded (/lib/systemd/system/zabbix-agent.service; enabled; vendor preset: enabled)
   Active: active (running) since Sun 2025-05-04 12:23:32 +0530; 31s ago
   Main PID: 22680 (zabbix_agentd)
     Tasks: 7 (limit: 4554)
    Memory: 6.2M
   CGroup: /system.slice/zabbix-agent.service
           └─22680 /usr/sbin/zabbix_agentd -c /etc/zabbix/zabbix_agentd.conf
             └─22688 /usr/sbin/zabbix_agentd: collector [idle 1 sec]
               └─22689 /usr/sbin/zabbix_agentd: listener #1 [waiting for connection]
                 └─22690 /usr/sbin/zabbix_agentd: listener #2 [waiting for connection]
                   └─22691 /usr/sbin/zabbix_agentd: listener #3 [waiting for connection]
                     └─22692 /usr/sbin/zabbix_agentd: active checks #1 [idle 1 sec]
                       └─22693 /usr/sbin/zabbix_agentd: active checks #2 [idle 1 sec]

May 04 12:23:31 dc.hewapathirana.sa systemd[1]: Starting Zabbix Agent...
May 04 12:23:32 dc.hewapathirana.sa systemd[1]: Started Zabbix Agent.
root@dc:~# sudo ufw allow 10050/tcp
Rules updated
Rules updated (v6)
root@dc:~#
```

3. Add Domain Controller in Zabbix Frontend

To add the Domain Controller to Zabbix, go to the Zabbix web interface and navigate to Configuration → Hosts. Click on Create host, then fill in the required fields: set the Hostname to domain-controller.local, assign it to a Group such as *Linux servers*, and under Interfaces, add an Agent interface using the Domain Controller's IP address. Next, switch to the Templates tab and link the host with a template like Template OS. Finally, click Add to save the host configuration.

New host

Host

IPMI

Tags

Macros

Inventory

Encryption

Value mapping

* Host name

dc.hewapathirana.sa

Visible name

Domain Controller

Templates

Linux by Zabbix agent

Select

* Groups

Linux servers

Select

Interfaces

Type	IP address	DNS name	Connect to	Port	Default
Agent	192.168.101.162		IP	DNS	10050

Add

Description

Monitored by proxy

(no proxy)

Enabled

Add

Cancel

4.6. Joining Fedora Client to Domain and Setting Up Zabbix Agent

1. Install Required Packages

```
root@csa-client:~  
File Edit View Search Terminal Help  
[root@csa-client ~]# sudo dnf install -y realmd sssd oddjob oddjob-mkhomedir adcli samba-common-tools krb5-workstation  
Last metadata expiration check: 0:28:30 ago on Sun 04 May 2025 10:35:53 AM +0530.  
Package realmd-0.16.3-19.fc29.x86_64 is already installed.  
Package sssd-2.2.0-3.fc29.x86_64 is already installed.  
Package oddjob-0.34.4-6.fc29.x86_64 is already installed.  
Package oddjob-mkhomedir-0.34.4-6.fc29.x86_64 is already installed.  
Package adcli-0.8.2-2.fc29.x86_64 is already installed.  
Dependencies resolved.  
=====
```

Package	Architecture	Version	Repository	Size
Installing:				
krb5-workstation	x86_64	1.16.1-26.fc29	updates	821 k
samba-common-tools	x86_64	2:4.9.15-0.fc29	updates	392 k
Installing dependencies:				
libkadm5	x86_64	1.16.1-26.fc29	updates	76 k

```
Transaction Summary  
=====
```

Transaction Summary	
Install	3 Packages

```
Total download size: 1.3 M  
Installed size: 4.6 M  
Downloading Packages:  
[=====] --- B/s | 0 B --- ETA
```

2. Discover the Domain

```
hewapathirana.sa  
[root@csa-client ~]# realm discover hewapathirana.sa  
hewapathirana.sa  
type: kerberos  
realm-name: HEWAPATHIRANA.SA  
domain-name: hewapathirana.sa  
configured: no  
server-software: active-directory  
client-software: sssd  
required-package: oddjob  
required-package: oddjob-mkhomedir  
required-package: sssd  
required-package: adcli  
required-package: samba-common-tools
```

3. Verify Domain Join and Verify Domain Join

`sudo realm join --user=Administrator YOUR.DOMAIN`

It will prompt for the domain admin password. If it is successful, now be part of the domain.

`realm list`

We can see the domain listed with the configured settings

```
root@csa-client
File Edit View Search Terminal Help
[root@csa-client ~]# sudo realm join --user=administrator hewapathirana.sa
Password for administrator:
[root@csa-client ~]# realm list
hewapathirana.sa
  type: kerberos
  realm-name: HEWAPATHIRANA.SA
  domain-name: hewapathirana.sa
  configured: kerberos-member
  server-software: active-directory
  client-software: sssd
  required-package: oddjob
  required-package: oddjob-mkhomedir
  required-package: sssd
  required-package: adcli
  required-package: samba-common-tools
  login-formats: %U@hewapathirana.sa
  login-policy: allow-realm-logins
[root@csa-client ~]#
```

Check using `su- administrator@hewapathirana.sa`

```
administrator@hewapathirana.sa@client1:~
File Edit View Search Terminal Tabs Help
root@csa-client:~
administrator@hewapathirana.sa@client1:~
[sawera@client1 ~]$ su - administrator@hewapathirana.sa
Password:
creating home directory for administrator@hewapathirana.sa.
[administrator@hewapathirana.sa@client1 ~]$
```

4. Configure the Agent

`sudo nano /etc/zabbix/zabbix_agentd.conf`

Update these lines:

- Server = 192.168.101.162
- ServerActive = 192.168.101.162
- Hostname=fedora-client1.hewapathirana.sa

```
# Number of pre-forked instances of zabbix_agentd that process passive checks.
# If set to 0, disables passive checks and the agent will not listen on any TCP port.
#
# Mandatory: no
# Range: 0-100
# Default:
# StartAgents=3

##### Active checks related

### Option: ServerActive
# List of comma delimited IP:port (or hostname:port) pairs of Zabbix servers and Zabbix proxies for active checks.
# If port is not specified, default port is used.
# IPv6 addresses must be enclosed in square brackets if port for that host is specified.
# If port is not specified, square brackets for IPv6 addresses are optional.
# If this parameter is not specified, active checks are disabled.
# Example: ServerActive=127.0.0.1:20051,zabbix.domain,[:1]:30051,::1,[12fc::1]
#
# Mandatory: no
# Default:
# ServerActive=

ServerActive=192.168.101.162

### Option: Hostname
# Unique, case sensitive hostname.
# Required for active checks and must match hostname as configured on the server.
# Value is acquired from HostnameItem if undefined.
#
# Mandatory: no
# Default:
# Hostname=

Hostname=client1.hewapathirana.sa
```

5. Start and Enable the Agent

```
[root@csa-client ~]# vi /etc/zabbix/zabbix_agentd.conf
[root@csa-client ~]# systemctl enable zabbix-agent.service
Created symlink /etc/systemd/system/multi-user.target.wants/zabbix-agent.service → /usr/lib/systemd/system/zabbix-agent.service.
[root@csa-client ~]# systemctl start zabbix-agent.service
[root@csa-client ~]# systemctl status zabbix-agent.service
● zabbix-agent.service - Zabbix Monitor Agent
   Loaded: loaded (/usr/lib/systemd/system/zabbix-agent.service; enabled; vendor preset: disabled)
   Active: active (running) since Sun 2025-05-04 12:45:45 +0530; 13s ago
     Main PID: 54049 (zabbix_agentd)
        Tasks: 6 (limit: 4652)
       Memory: 3.7M
      CGroup: /system.slice/zabbix-agent.service
              └─54049 /usr/sbin/zabbix_agentd -f
                └─54050 /usr/sbin/zabbix_agentd: collector [idle 1 sec]
                  └─54051 /usr/sbin/zabbix_agentd: listener #1 [waiting for connection]
                    └─54052 /usr/sbin/zabbix_agentd: listener #2 [waiting for connection]
                      └─54053 /usr/sbin/zabbix_agentd: listener #3 [waiting for connection]
                        └─54054 /usr/sbin/zabbix_agentd: active checks #1 [idle 1 sec]

May 04 12:45:45 client1.hewapathirana.sa systemd[1]: Started Zabbix Monitor Agent.
May 04 12:45:45 client1.hewapathirana.sa zabbix_agentd[54049]: Starting Zabbix Agent [client1.hewapathirana.sa]. Zabbix 3.0.28 (revision 0867a35b15).
May 04 12:45:45 client1.hewapathirana.sa zabbix_agentd[54049]: Press Ctrl+C to exit.
[root@csa-client ~]#
```

6. Add Fedora Client to Zabbix Web Interface

To add the Fedora client to Zabbix, go to the Zabbix frontend, navigate to Configuration → Hosts, and click on Create host. In the host configuration form, set the Hostname to `fedora-client.local`, assign it to a Group such as *Linux servers* or any custom group, and under Interfaces, add an Agent interface using the Fedora client's IP address. Then, go to the Templates tab and link the host with Template OS Linux to enable basic monitoring. Finally, click Add to complete the setup.

New host

Host

IPMI

Tags

Macros

Inventory

Encryption

Value mapping

* Host name

client1.hewapathirana.sa

Visible name

Client

Templates

Linux by Zabbix agent

type here to search

Select

* Groups

Linux servers

type here to search

Select

Interfaces

Type	IP address	DNS name	Connect to	Port	Default
Agent	192.168.101.158		IP	DNS	10050

Add

Description

Monitored by proxy

(no proxy)

Enabled

☒

Add

Cancel

7. Created Hosts List

1. Client

- IP: 192.168.101.158:10050
- Zabbix agent is available (ZBX green)

2. Domain Controller

- IP: 192.168.101.162:10050
- Zabbix agent is available (ZBX green)

3. Zabbix server

- IP: 127.0.0.1:10050 (localhost)
- Zabbix agent is available (ZBX green)

The screenshot displays the Zabbix 6.0.40 interface for managing hosts. On the left is a sidebar with navigation links: Monitoring, Dashboard, Problems, Hosts, Latest data, Maps, Discovery, Services, Inventory, Reports, Configuration, Administration, Support, Integrations, Help, User settings, and Sign out. The main content area is titled 'Hosts' and includes a 'Create host' button. Below the title is a form for adding new hosts with fields for Name, Host groups, IP, DNS, Port, Status (Any, Enabled, Disabled), Tags (And/Or, Or), and Severity (Not classified, Warning, High, Information, Average, Disaster). There are also checkboxes for 'Show hosts in maintenance' and 'Show suppressed problems'. Below the form is a table of existing hosts.

Name	Interface	Availability	Tags	Status	Latest data	Problems	Graphs	Dashboards	Web
Client	192.168.101.158:10050	ZBX	class: os target: linux	Enabled	Latest data 42	1	Graphs 8	Dashboards 2	Web
Domain Controller	192.168.101.162:10050	ZBX	class: os target: linux	Enabled	Latest data 68	1	Graphs 14	Dashboards 2	Web
Zabbix server	127.0.0.1:10050	ZBX	class: os class: software target: linux ***	Enabled	Latest data 125	1	Graphs 30	Dashboards 3	Web

Displaying 3 of 3 found

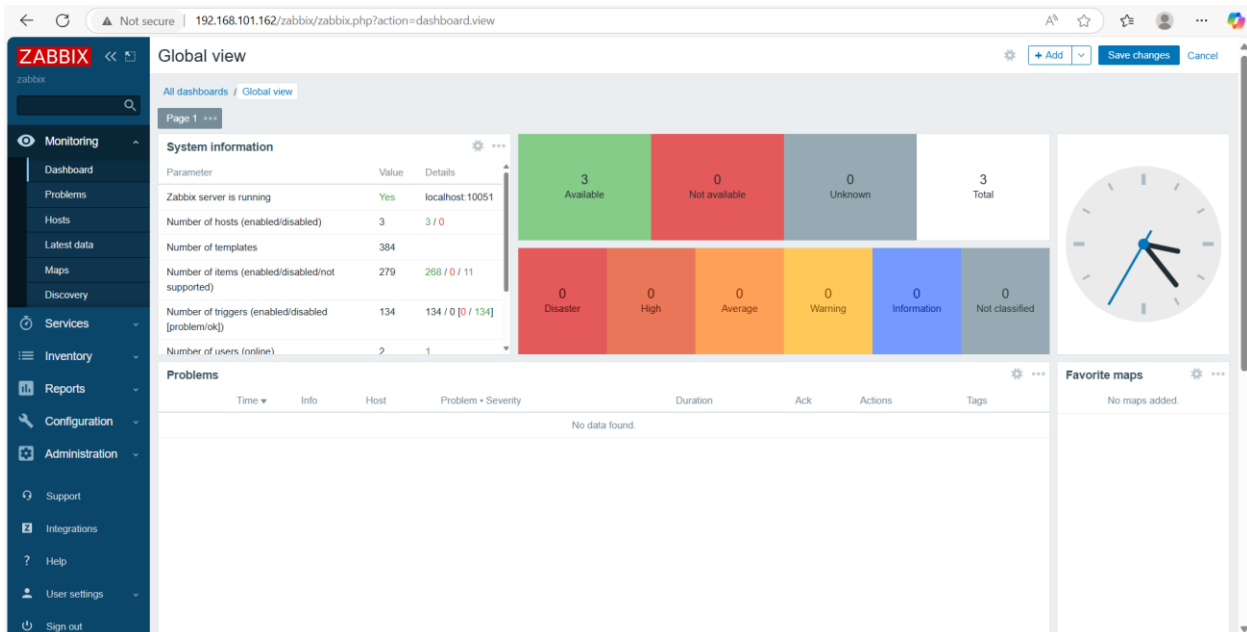
Zabbix 6.0.40. © 2001–2025, Zabbix SIA

5. Monitoring Setup

In the monitoring setup phase, the goal is to ensure that both the Linux Domain Controller and the Fedora client are being accurately monitored through the Zabbix server. This involves linking hosts with relevant templates, verifying data collection, and organizing dashboards for clear visibility.

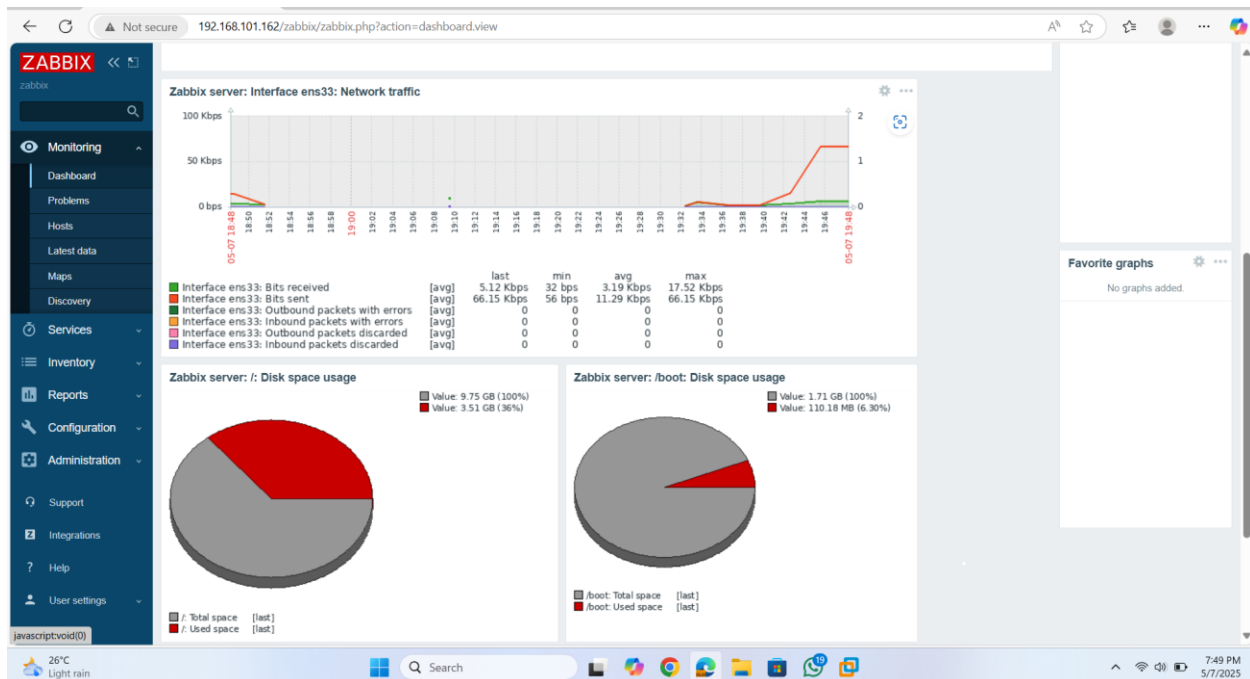
Once Zabbix agents are properly configured on both the domain controller and client system, and relevant monitoring templates are applied, the Zabbix dashboard starts displaying real-time and historical data

1. Zabbix Dashboard



I've added that the Zabbix dashboard is essential and practical for basic monitoring, for example

1. **Network Traffic Graph (Interface ens33)** – Helps you track real-time bandwidth usage, identify abnormal spikes, and ensure network interfaces are functioning properly.
2. **Disk Space Usage Pie Charts** – Give a quick visual of storage status, allowing you to spot potential issues with root or boot partitions before they become critical.



These widgets give instant insights into resource utilization without having to manually check logs or run commands. Such graphs help detect anomalies, plan capacity upgrades, and maintain performance stability across systems. This setup is especially critical in production environments where continuous availability is essential.

6. Challenges and Solutions

1. Agent not showing data

Ensure firewall ports are open (default: TCP 10050) and that the agent's config matches the hostname in Zabbix.

```
[root@csa-client ~]# firewall-cmd --add-port=10050/tcp --permanent
success
[root@csa-client ~]# firewall-cmd --reload
success
[root@csa-client ~]#
```

Verify that the Server and Hostname parameters in `/etc/zabbix/zabbix_agentd.conf` match the IP of the Zabbix server and the exact hostname you've added in the frontend

2. Time zone mismatch or graph delays

Set the matching time zone in the PHP config and Zabbix frontend.

```
sawera@ubuntu:~$ sudo timedatectl set-timezone Asia/Karachi
sawera@ubuntu:~$ timedatectl
          Local time: Tue 2025-04-08 13:49:31 PKT
          Universal time: Tue 2025-04-08 08:49:31 UTC
             RTC time: Tue 2025-04-08 08:49:31
            Time zone: Asia/Karachi (PKT, +0500)
System clock synchronized: yes
              NTP service: active
          RTC in local TZ: no
sawera@ubuntu:~$ sudo timedatectl set-timezone Asia/Colombo
sawera@ubuntu:~$ timedatectl
          Local time: Tue 2025-04-08 14:23:18 +0530
          Universal time: Tue 2025-04-08 08:53:18 UTC
             RTC time: Tue 2025-04-08 08:53:19
            Time zone: Asia/Colombo (+0530, +0530)
System clock synchronized: yes
              NTP service: active
          RTC in local TZ: no
sawera@ubuntu:~$
```

Attempts to join the Fedora client to the Samba AD domain might fail due to unresolved domain names or missing tools.

- ## Vi /etc /resolv.conf

40

4. Agent not starting automatically

After installation, the Zabbix agent might not start or might not stay running across reboots.

Enable the service to auto-start: `sudo systemctl enable zabbix-agent`

Start it manually: `sudo systemctl start zabbix-agent`

Check the agent's status: `sudo systemctl status zabbix-agent`

If the service fails, check logs for errors: `sudo cat /var/log/zabbix/zabbix_agentd.log`

Common issues include syntax errors in the configuration file, incorrect server IP, or missing permissions.

7. Conclusion

This project successfully demonstrated the implementation of a **Linux-based Domain Controller** using **Samba with Active Directory (AD)** support, and the integration of **Zabbix monitoring** for both the server and a Fedora-based client machine. The process involved configuring Samba to function in **AD DC mode**, enabling centralized authentication and domain services within a Linux environment.

A significant part of the project included the deployment of **Zabbix Server**, a powerful open-source monitoring platform. The Zabbix server was configured with **PostgreSQL as the backend database**, Apache as the web server, and PHP for the frontend. Monitoring was extended to both the Domain Controller and the Fedora client using **Zabbix Agents**, which allowed for continuous tracking of system performance, availability, and health metrics.

The **agent installation and host registration** in the Zabbix frontend were followed by the application of **OS-level monitoring templates** for both the server and client, enabling automated data collection and real-time visualization through graphs, alerts, and logs.

Several **challenges** were encountered, including agent communication issues, time zone mismatches, domain joining failures on Fedora, and agent service startup problems. Each issue was addressed through systematic troubleshooting, proper configuration, and the use of relevant Linux tools (e.g., firewall-cmd, realmd, systemctl, ntpd). These solutions not only ensured system stability but also improved understanding of service dependencies in enterprise-level Linux environments.

In conclusion, this project highlights how open-source tools like **Samba** and **Zabbix** can be leveraged to create a cost-effective, scalable, and efficient **network and system management infrastructure**. It demonstrates a practical approach to real-world IT challenges in authentication, domain control, and proactive monitoring—skills highly relevant in enterprise environments. This setup can be expanded in the future to include **custom triggers, advanced templates, email/SMS notifications, dashboard customization**, and monitoring of additional devices and services across the network.

8. References

1. **Zabbix Documentation (Official)**

<https://www.zabbix.com/documentation/current>

Official documentation for installing, configuring, and using Zabbix, including frontend, agents, and templates.

2. **Samba Wiki – Setting up Samba as an Active Directory Domain Controller**

https://wiki.samba.org/index.php/Setting_up_Samba_as_an_Active_Directory_Domain_Controller

Step-by-step guide on configuring Samba AD DC on Linux.

3. **Ubuntu Server Guide**

<https://ubuntu.com/server/docs>

Includes system setup, Samba installation, and security configurations.

4. **Zabbix YouTube Channel**

<https://www.youtube.com/c/Zabbix>

Video tutorials for Zabbix server, agent setup, dashboard customization, and advanced features.

5. **Zabbix Share (Community Templates & Dashboards)**

<https://share.zabbix.com/>

A repository of user-contributed templates, dashboards, scripts, and modules for Zabbix.